

Topic 1: States of matter

Premium Answer Booklet

6 questions	2020-2025 Paper 4	Premium readable layout
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Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

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Question 01

11 marks

2020 May/June Paper 42 Question 1

Main topic: Topic 1: States of matter

Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 8: The Periodic Table, Topic 9: Metals, Topic 4: Electrochemistry, Topic 7:

Answer source label: 0620_s20_ms_42.pdf

1(a)(i)	• condensation
1(a)(ii)	• sublimation
1(a)(iii)	• fractional distillation
1(a)(iv)	• electrolysis
1(a)(v)	• chromatography
1(b)(i)	• N
1(b)(ii)	• O
1(b)(iii)	• Be
1(b)(iv)	• N
1(b)(v)	• Ne
1(b)(vi)	• Li

Easy explanation / reminder

- Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 02

6 marks

2020 May/June Paper 43 Question 5

Main topic: Topic 1: States of matter
Cross-topic links: Topic 7: Acids, bases and salts
Answer source label: 0620_s20_ms_43.pdf

5(a)	• particle • separation • particle • arrangement • type of • motion • solid • touching • liquid • random • gas • random
5(b)	• condensing
5(c)	• evaporation • diffusion

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 03

10 marks

2021 Feb/March Paper 42 Question 2

Main topic: Topic 1: States of matter
Cross-topic links: Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds
Answer source label: 0620_m21_ms_42.pdf

2(a)	• chlorine
2(b)	• fluorine
2(c)(i)	• random motion of molecules / particles
2(c)(ii)	• hydrogen • lowest (relative) molecular mass
2(d)(i)	• No separate point listed.
2(d)(ii)	• argon / Ar
2(d)(iii)	• carbon dioxide
2(d)(iv)	• fractional (1) • distillation (1)

Easy explanation / reminder

- Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates
- clearly stated.

END OF ANSWER

Question 04

11 marks

2021 Oct/Nov Paper 42 Question 1

Main topic: Topic 1: States of matter
Answer source label: 0620_w21_ms_42.pdf

1(a)	• 1 mark for each correct row • State • touching • random • movement • regularly • arranged • Gas • ✓ • Liquid • ✓ • ✓ • Solid • ✓ • ✓
1(b)(i)	• boiling happens at a specific temperature (1) • boiling has bubbles (1)
1(b)(ii)	• condensation (1) • sublimation (1)
1(c)	• one horizontal line level with Y (1) • two separate decreases before and after horizontal line (1)
1(d)(i)	• dissolving
1(d)(ii)	• precipitation

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
- reduced by carbon/CO.

END OF ANSWER

Question 05

10 marks

2022 Oct/Nov Paper 43 Question 2

Main topic: Topic 1: States of matter
Cross-topic links: Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry
Answer source label: 0620_w22_ms_43.pdf

2(a)(i)	• E
2(a)(ii)	• H
2(a)(iii)	• G
2(b)	• D (1) • conducts electricity when solid (1)
2(c)	• F (1) • high melting point (1) • non-conductor of electricity when solid and liquid (1)
2(d)	• I (1) • conducts electricity when liquid but not when solid / ONLY conducts when liquid (1)

END OF ANSWER

Question 06

13 marks

2025 Oct/Nov Paper 42 Question 1

Main topic: Topic 1: States of matter
Cross-topic links: Topic 10: Chemistry of the environment
Answer source label: 0620_w25_ms_42.pdf

1(a)	• particle separation • particle • arrangement • particle motion • solid • uniform • vibrate • liquid • touching • random • random • gas • apart
1(b)	• melting • condensing / condensation • dissolving
1(c)(i)	• motion / movement of particles
1(c)(ii)	• relative molecular mass
1(d)	• photochemical smog • respiratory problems
1(e)(i)	• carbon monoxide
1(e)(ii)	• nitrogen • carbon dioxide

Easy explanation / reminder

- Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
- Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.

END OF ANSWER